

Jamia Millia Islamia

A Central University

DSC-503: COMPUTER NETWORKS

MR. HANNAN MANSOOR

DEPARTMENT OF COMPUTER ENGINEERING

JAMIA MILLIA ISLAMIA

UNIT- 3: THE MEDIUM ACCESS CONTROL SUB LAYER

- The channel allocation problem,
- ALOHA,
- Multiple access Protocols,
- Collision free Protocols,
- IEEE Standards for LANs and MANs,
- Wireless LAN: IEEE 802.11,
- High speed LANs.

© HANNAN MANSOOR

Data Link Layer Services

- Framing
- Link Access
- Reliable Delivery
- Error Detection and Correction
- Flow Control

© HANNAN MANSOOR

NIC

- Data link layer implemented on-chip or in network interface card (NIC).
- NIC implements link, physical layer.
- It is attached into host's system buses.
- Combination of hardware, software, firmware.

© HANNAN MANSOOR

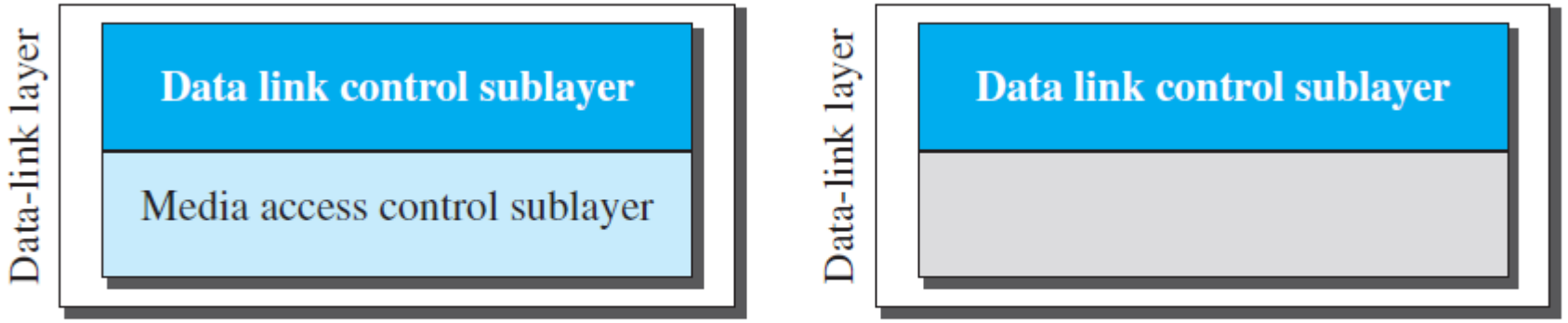
Multiple Access Links, Protocols

- Two types of “links”
  - Point-to-point
  - Point to point link between Ethernet switch, host.
  - PPP for dial-up access.
- Broadcast (shared wire or medium)
  - Old-school Ethernet
  - Upstream HFC in cable –based access network
  - 802.11 wireless LAN, 4G/5G, satellite.

© HANNAN MANSOOR

DATA LINK LAYER: TWO SUBLAYER

- Data link control (DLC) and
- Media access control (MAC).



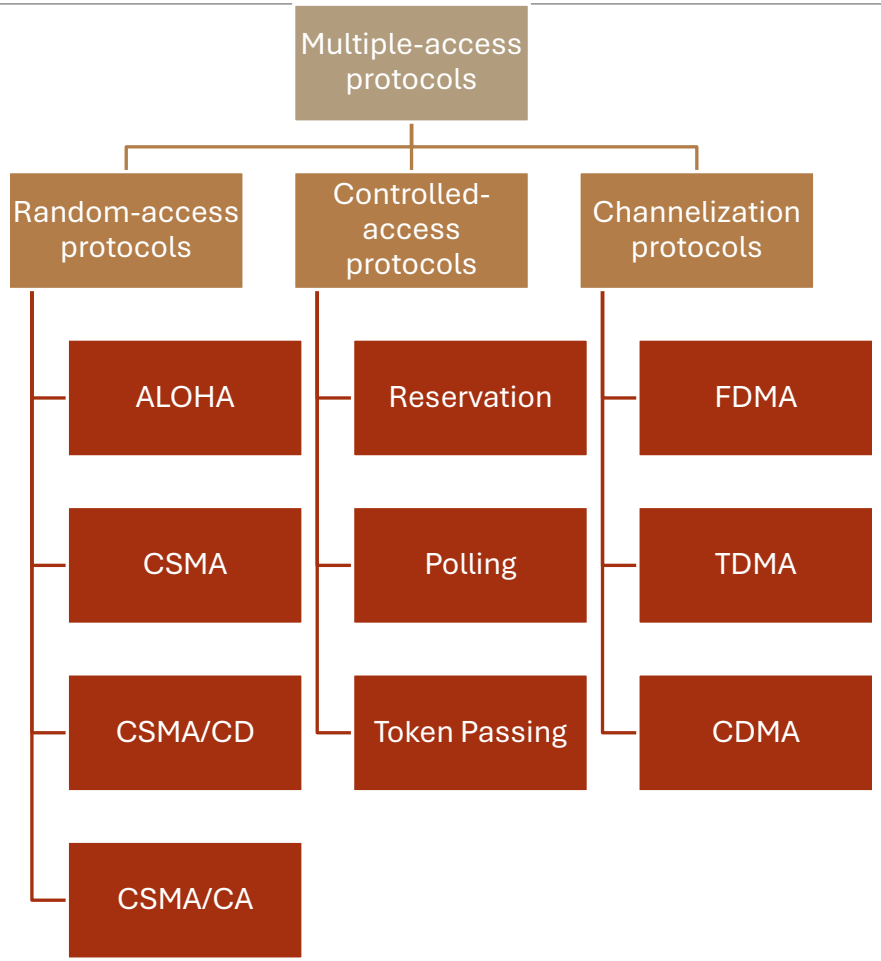
a. Data-link layer of a broadcast link

b. Data-link layer of a point-to-point link

Media Access Control: Introduction

- When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.
- Many formal protocols have been devised to handle access to a shared link.

Taxonomy of multiple-access protocols



© HANNAN MANSOOR

Random Access

ALOHA

CSMA

CSMA/CD

CSMA/CA

© HANNAN MANSOOR

RANDOM ACCESS

- In random-access or contention methods,
  - no station is superior to another station and
  - none is assigned control over another.
  - No station permits, or does not permit, another station to send.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
  - This decision depends on the state of the medium (idle or busy).
- In other words, each station can transmit when it desires on the condition that it follows the predefined procedure, including testing the state of the medium.

© HANNAN MANSOOR

Features of Random Access

- Two features give this method its name.
- First, there is no scheduled time for a station to transmit.
  - Transmission is random among the stations.
  - That is why these methods are called random access.
- Second, no rules specify which station should send next.
  - Stations compete with one another to access the medium.
  - That is why these methods are also called contention methods
- In a random-access method, each station has the right to the medium without being controlled by any other station.
- However, if more than one station tries to send, there is an access conflict—collision—and the frames will be either destroyed or modified.

Points to Ponder for Multiple access

- To avoid access conflict or to resolve it when it happens, each station follows a procedure that answers the following questions:
  - When can the station access the medium?
  - What can the station do if the medium is busy?
  - How can the station determine the success or failure of the transmission?
  - What can the station do if there is an access conflict?

© HANNAN MANSOOR

ALOHA

- ALOHA, the earliest random access method, was developed at the University of Hawaii in early 1970.
- It was designed for a radio (wireless) LAN, but it can be used on any shared medium.
- It is obvious that there are potential collisions in this arrangement.
- The medium is shared between the stations.
- When a station sends data, another station may attempt to do so at the same time.
- The data from the two stations collide and become garbled.

© HANNAN MANSOOR

Pure ALOHA

- The original ALOHA protocol is called pure ALOHA.
- This is a simple, but elegant protocol.
- The idea is that each station sends a frame whenever it has a frame to send.
- However, since there is only one channel to share, there is the possibility of collision between frames from different stations.

© HANNAN MANSOOR

Figure: *Frames in a pure ALOHA network*

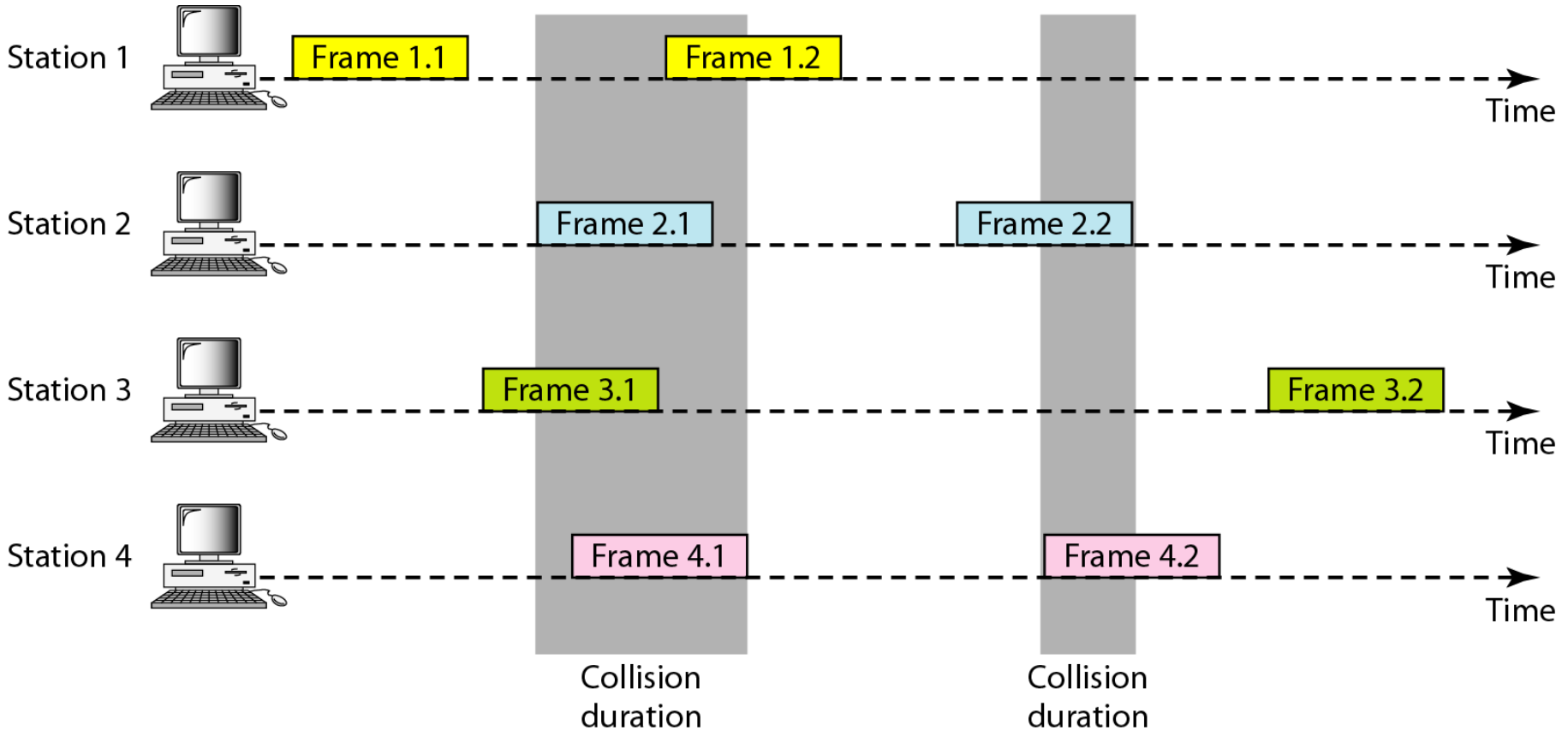


Figure: shows an example of frame collisions in pure ALOHA.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

© HANNAN MANSOOR

Pure ALOHA Working

- The pure ALOHA protocol relies on acknowledgments from the receiver.
- When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive after a time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame.
- The randomness will help avoid more collisions.
- We call this time the back-off time T_B.
- Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames.
- After a maximum number of retransmission attempts K_{max} a station must give up and try later.

© HANNAN MANSOOR

Procedure for pure ALOHA protocol

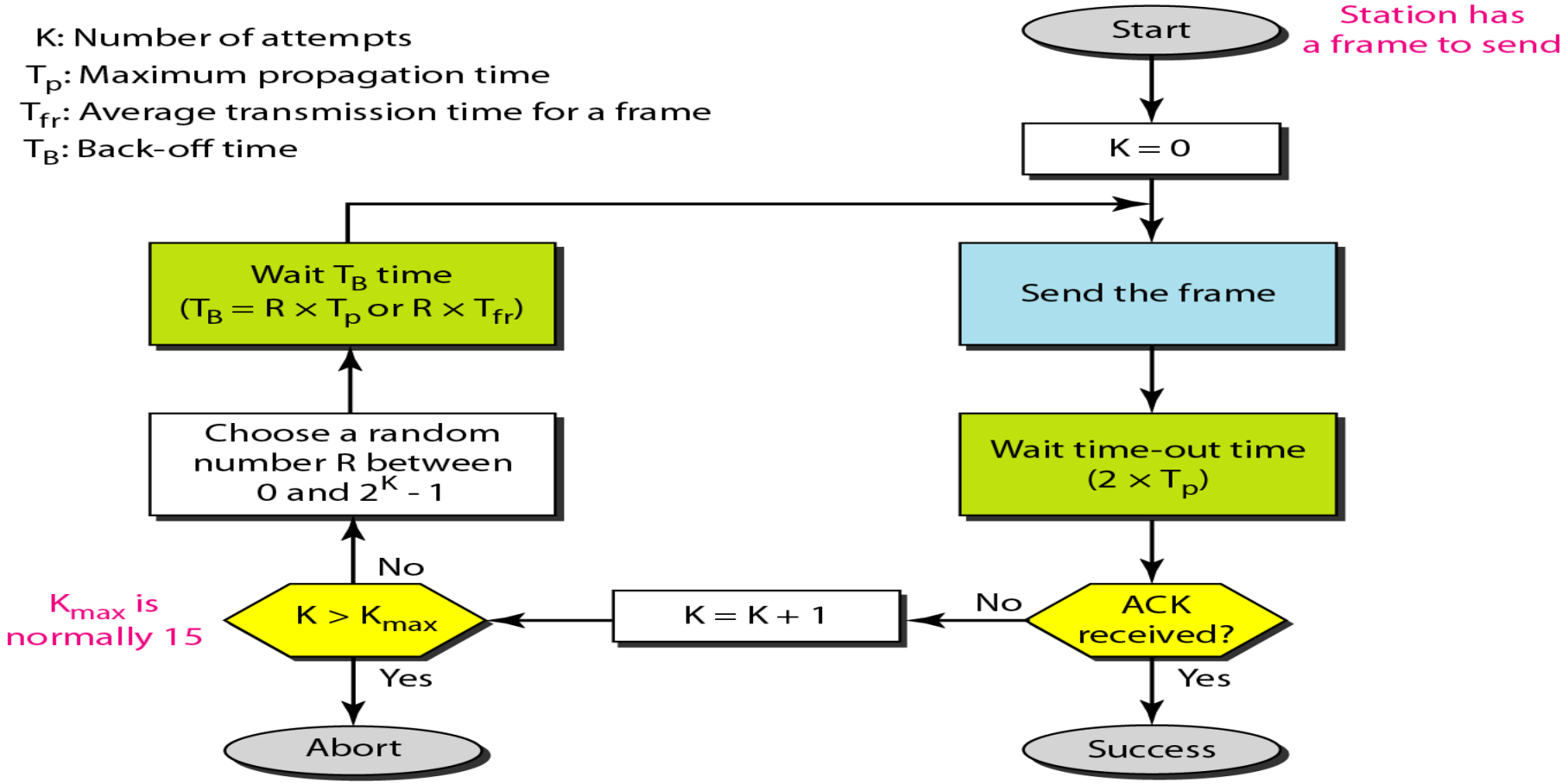


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Timings in Pure ALOHA

- The time-out period is equal to the maximum possible round-trip propagation delay,
  - which is twice the amount of time required to send a frame between the two most widely separated stations $(2 \times T_p)$.
  - The back-off time T_B is a random value that normally depends on K (the number of attempted unsuccessful transmissions).
  - The formula for T_B depends on the implementation.

© HANNAN MANSOOR

Binary exponential back-off.

- One common formula is the binary exponential back-off.
- In this method, for each retransmission, a multiplier in the range $\mathrm{R} = 0$ to $2^K - 1$ is randomly chosen and multiplied by T_p (maximum propagation time) or T_{fr} (the average time required to send out a frame) to find T_B .
- Note that in this procedure, the range of the random numbers increases after each collision.
- The value of K_{max} is usually chosen as 15.

© HANNAN MANSOOR

Example 1

- The stations on a wireless ALOHA network are a maximum of *600* km apart. If we assume that signals propagate at $3\times10^{8} m/s$, we find

$$T_p = \frac{(600 \times 10^3)}{(3 \times 10^8)} = 2m/s.$$

Now find the value of T_B for different values of K.

© HANNAN MANSOOR

Solution

- a. For K = 1, the range is {0, 1}. The station needs to generate a random number with a value of 0 or 1. This means that T_B is either 0 ms (0 × 2) or 2 ms (1 × 2), based on the outcome of the random variable.
- b. For K = 2, the range is {0, 1, 2, 3}. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable.

Solution

- c. For K = 3, the range is {0, 1, 2, 3, 4, 5, 6, 7}. This means that T_B can be 0, 2, 4, . . . , 14 ms, based on the outcome of the random variable.

- d. We need to mention that if K > 10, it is normally set to 10.

Example 2:

The stations on a wireless ALOHA network are a maximum of 600 km apart.

If we assume that signals propagate at

$3\times10^{8} m/s$, we find

$$T_p = \frac{(600 \times 10^3)}{(3 \times 10^8)} = 2m/s.$$

For K = 2, the range of R is {0, 1, 2, 3}.

This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable R.

Pure ALOHA Protocol Efficiency

- $S = Ge^{-2G}$
  - Where:
  - S = Throughput (efficiency)
  - G = Average number of transmission attempts per frame time
  - e = Euler's constant
- To find the **maximum efficiency**, differentiate $S = Ge^{-2G}$ and set it to zero.
- The maximum occurs at: $G = \frac{1}{2}$

$$S_{max} = \frac{1}{2}e^{-1}$$

$$S_{max} = \frac{1}{2 \times 2.718}$$

$$S_{max} \approx 0.184$$

- So, the **maximum efficiency of PURE ALOHA is: 18.4%**

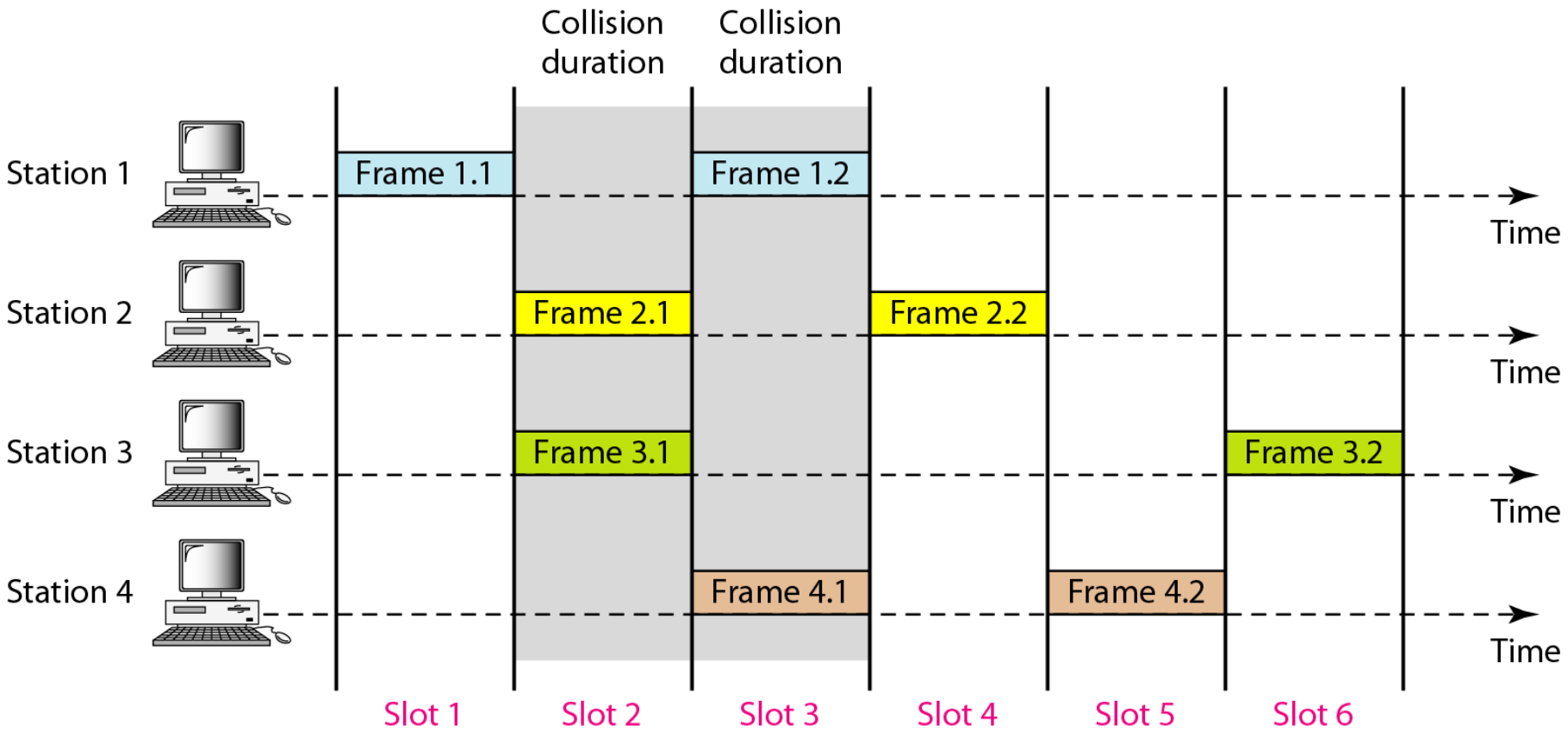
Image adapted From: "\"

Slotted ALOHA

- Pure ALOHA has a vulnerable time of 2 × Tfr. This is so because there is no rule that defines when the station can send.
- A station may send soon after another station has started or just before another station has finished.
- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- In slotted ALOHA we divide the time into slots of Tfr seconds and force the station to send only at the beginning of the time slot.

© HANNAN MANSOOR

Frames in a slotted ALOHA network



© HANNAN MANSOOR

Slotted ALOHA: Efficiency

efficiency: long-run fraction of successful slots (many nodes, all with many frames to send)

- *suppose:* N nodes with many frames to send, each transmits in slot with probability p
  - prob that given node has success in a slot $= p(1-p)^{N-1}$
  - prob that *any* node has a success $= Np(1-p)^{N-1}$
  - max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$
  - for many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives:
- *max efficiency = 1/e = .37*
- *at best:* channel used for useful transmissions 37% of time!

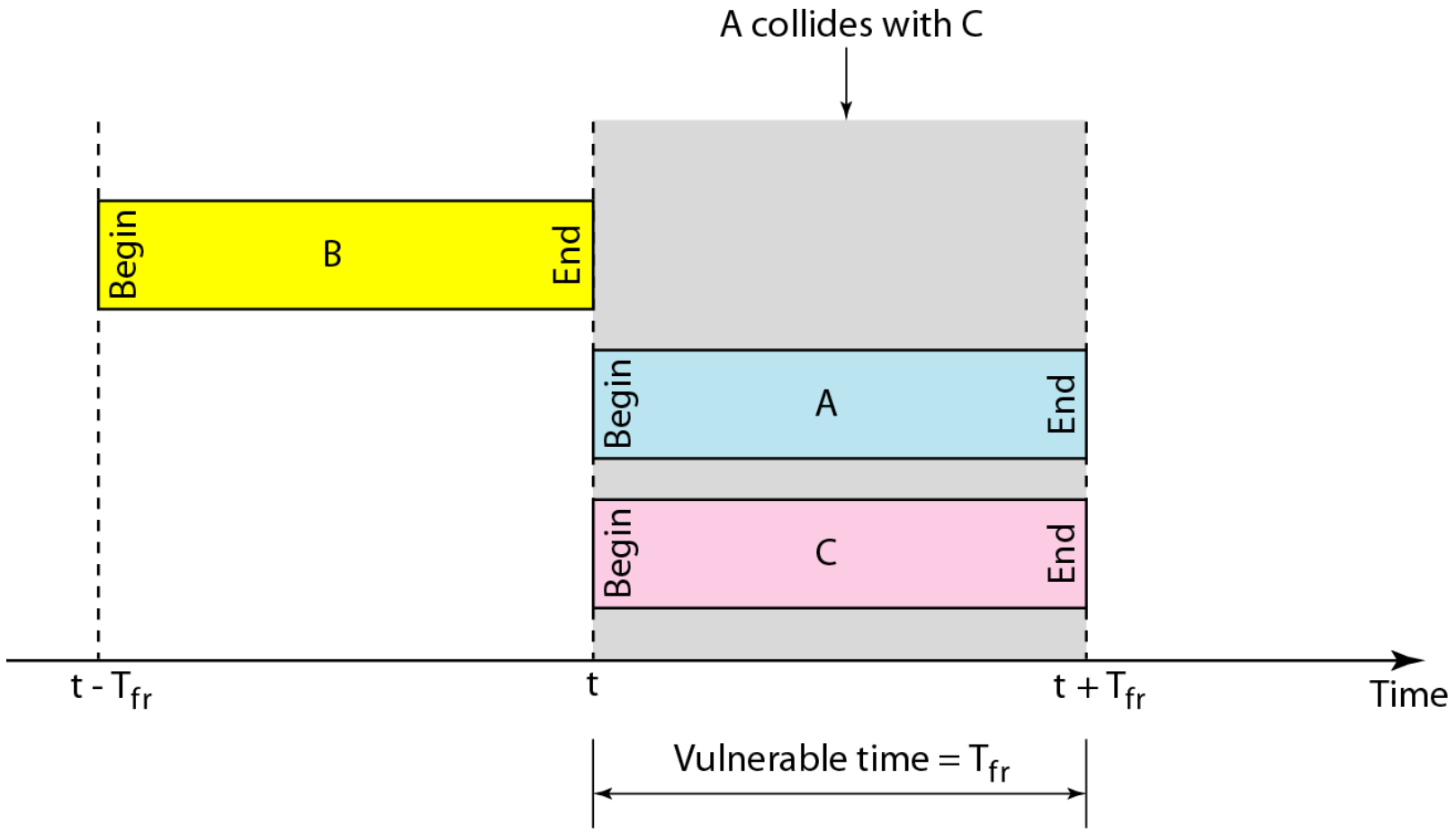
The throughput for slotted ALOHA is
$S = G \times e^{-G}$.
The maximum throughput
$S_{max} = 0.368$ **when G = 1.**

Slotted ALOHA: Efficiency

The throughput for slotted ALOHA is
$S = G \times e^{-G}$.
The maximum throughput
$S_{max} = 0.368$ **when G = 1.**

- **efficiency**: long-run fraction of successful slots (many nodes, all with many frames to send)
- **suppose**: Nnodes with many frames to send, each transmits in slot with probability p
- **prob that given node has success in a slot** $= p(1-p)^{N-1}$
- **prob that any node has a success** $= Np(1-p)^{N-1}$
- **max efficiency**: find p^*that maximizes $Np(1-p)^{N-1}$
- **for many nodes**, take limit of $Np^*(1-p^*)^{N-1}$as Ngoes to infinity, gives:
- **max efficiency** $= 1/e = 0.37$
- **at best**: channel used for useful transmissions $37\%$of time!

Figure *Vulnerable time for slotted ALOHA protocol*



Example

A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

a. 1000 frames per second b. 500 frames per second

c. 250 frames per second.

Solution

The frame transmission time is 200/200 kbps or 1 ms.

a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1. In this case $S = G \times e^{-G}$ *or* $S = 0.368$ *(36.8 percent). This means that the throughput is* $1000 \times 0.0368 = 368$ *frames. Only 386 frames out of 1000 will probably survive.*

Example (continued)

b. *If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2). In this case* $S = G \times e^{-G}$ *or* $S = 0.303$ *(30.3 percent). This means that the throughput is* $500 \times 0.0303 = 151$. *Only 151 frames out of 500 will probably survive.*

c. *If the system creates 250 frames per second, this is (1/4) frame per millisecond. The load is (1/4). In this case* $S = G \times e^{-G}$ *or* $S = 0.195$ *(19.5 percent). This means that the throughput is* $250 \times 0.195 = 49$. *Only 49 frames out of 250 will probably survive.*

Figure 12.8 *Space/time model of the collision in CSMA*

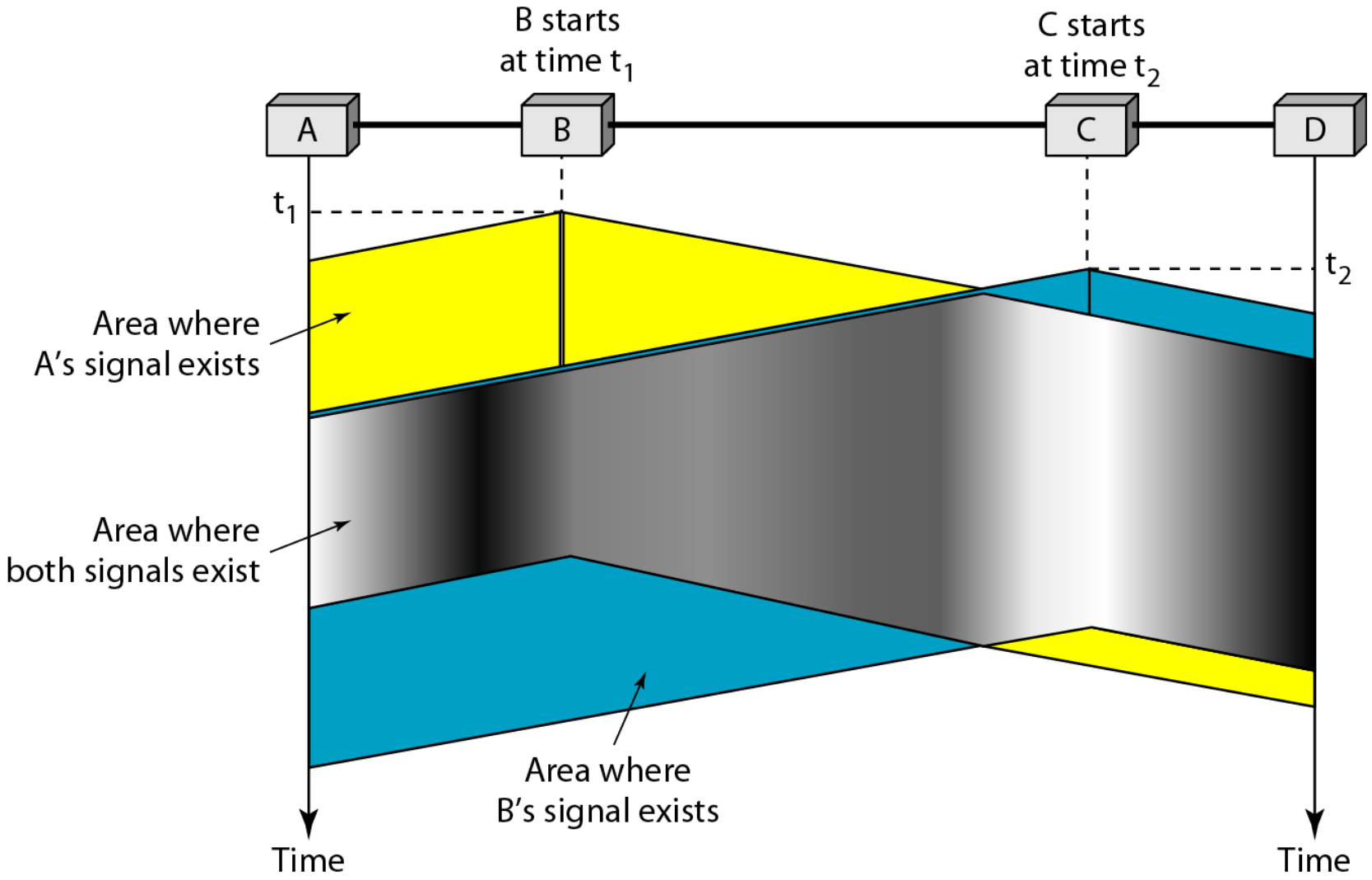


Figure 12.9 *Vulnerable time in CSMA*

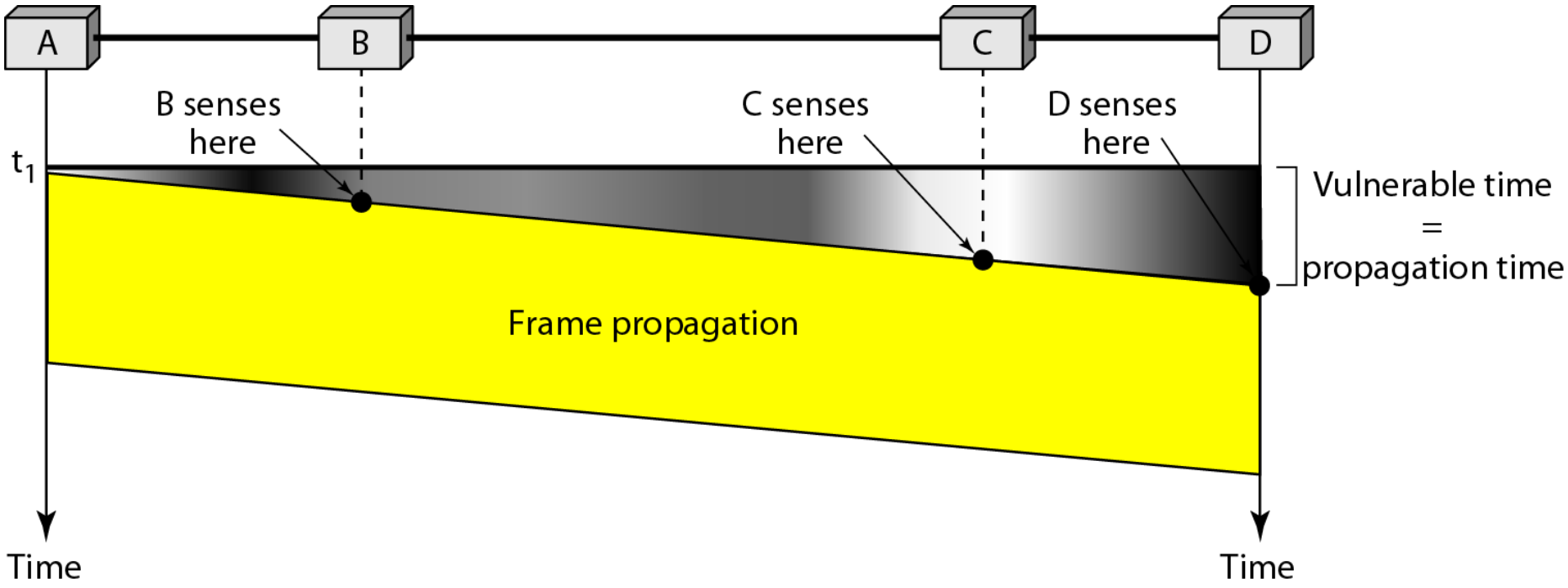
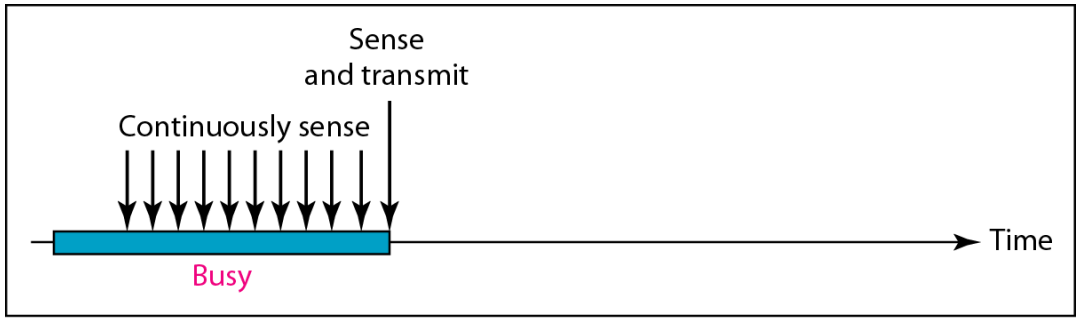
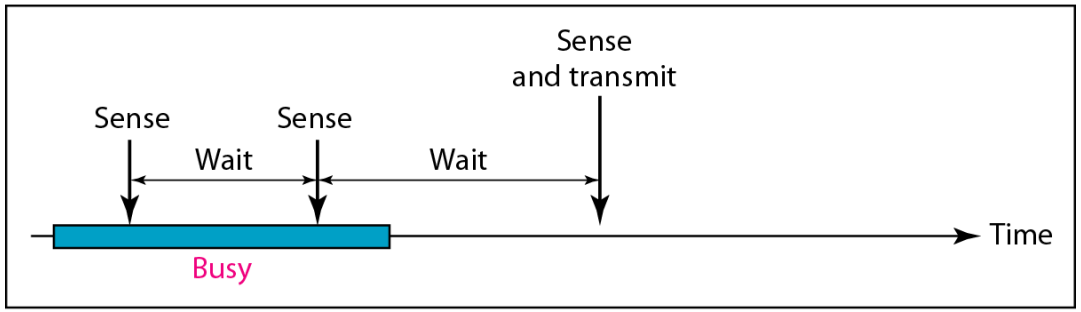


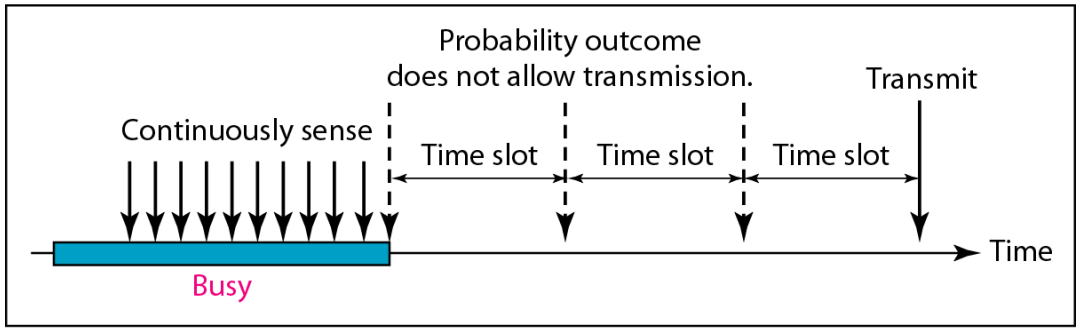
Figure 12.10 *Behavior of three persistence methods*



a. 1-persistent

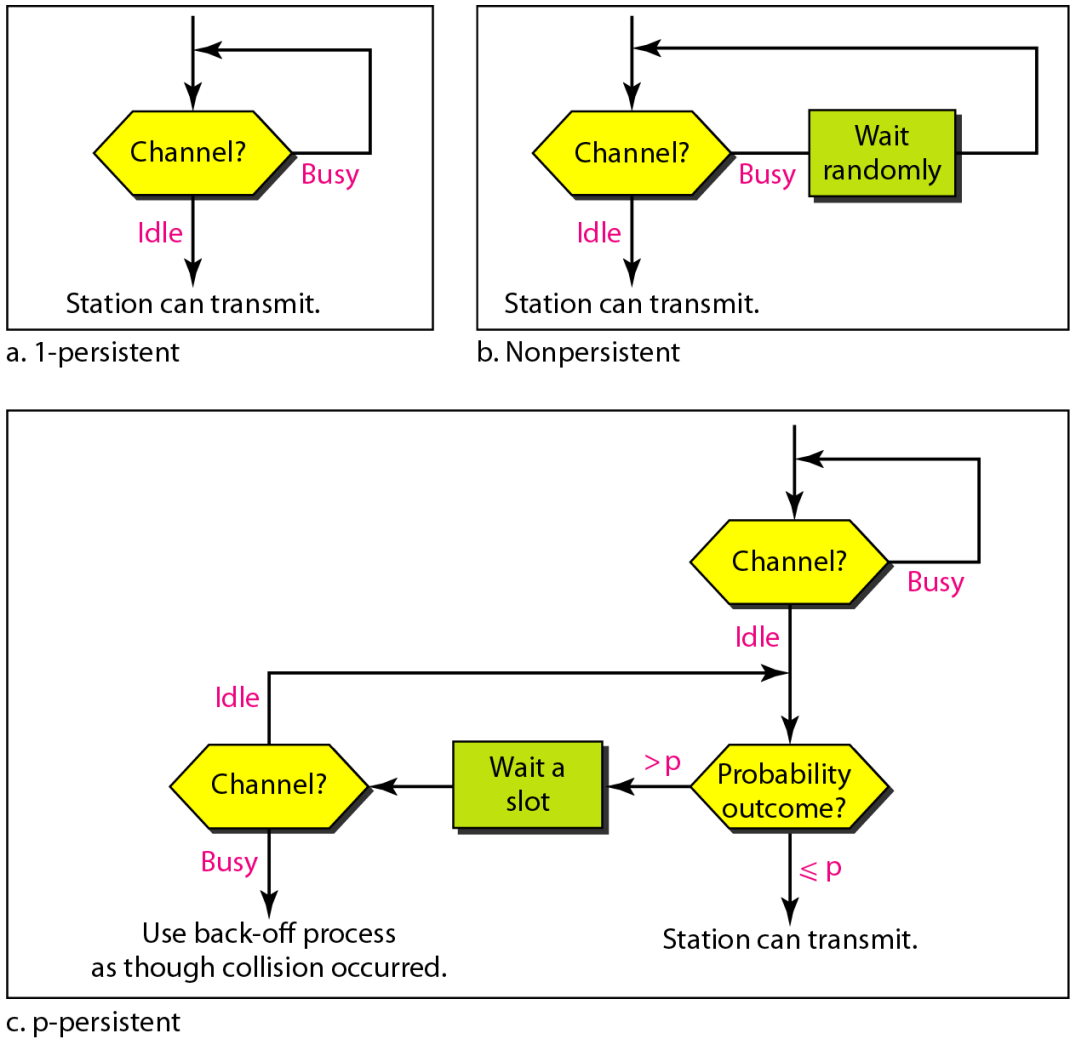


b. Nonpersistent



c. p-persistent

Figure 12.11 *Flow diagram for three persistence methods*



a. 1-persistent

b. Nonpersistent

c. p-persistent

Figure 12.12 *Collision of the first bit in CSMA/CD*

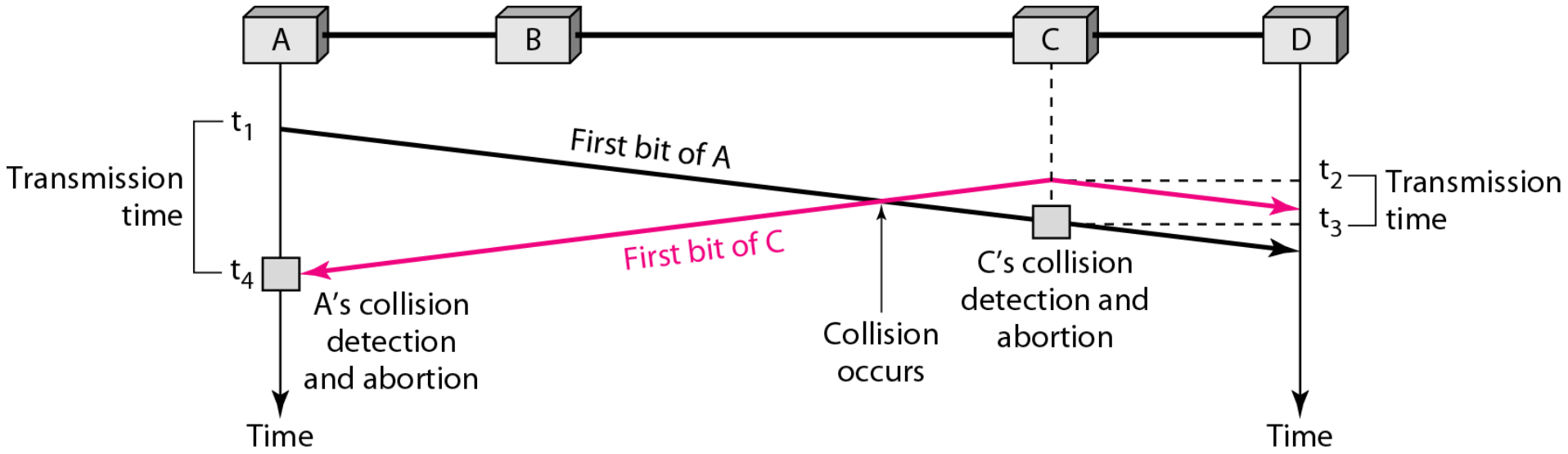
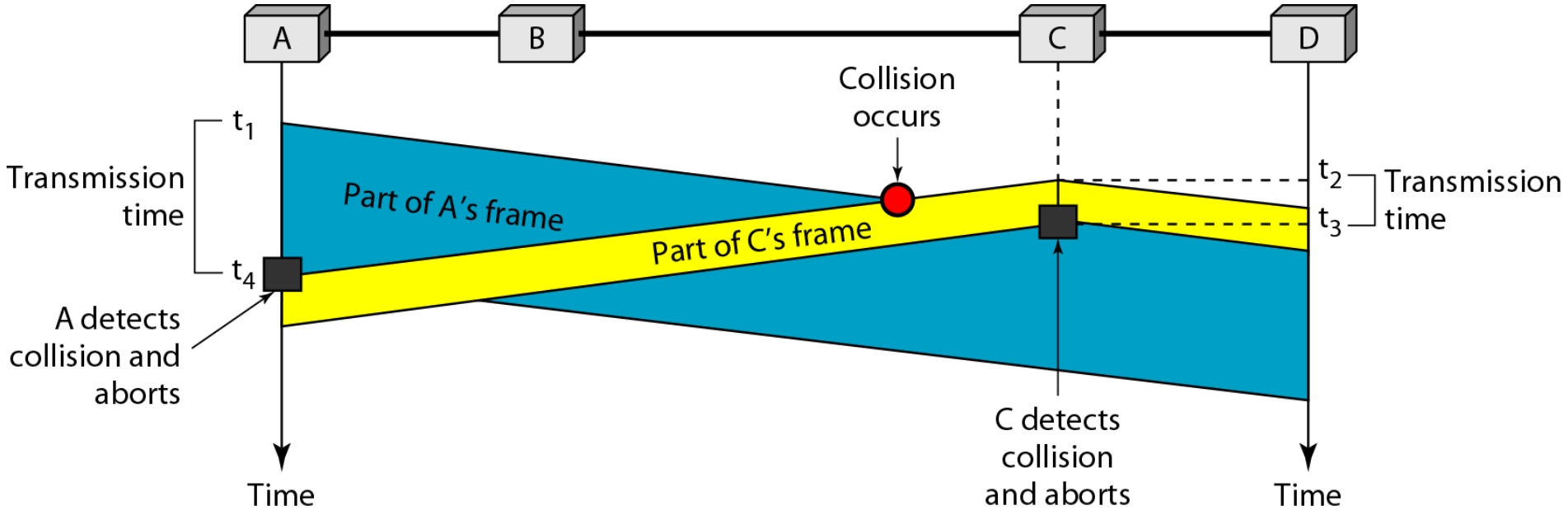


Figure 12.13 *Collision and abortion in CSMA/CD*



Example 12.5

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is 25.6 μs, what is the minimum size of the frame?

Solution

The frame transmission time is $T_{fr} = 2 \times T_p = 51.2$ μs. This means, in the worst case, a station needs to transmit for a period of 51.2 μs to detect the collision. The minimum size of the frame is 10 Mbps × 51.2 μs = 512 bits or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet.

Figure 12.14 *Flow diagram for the CSMA/CD*

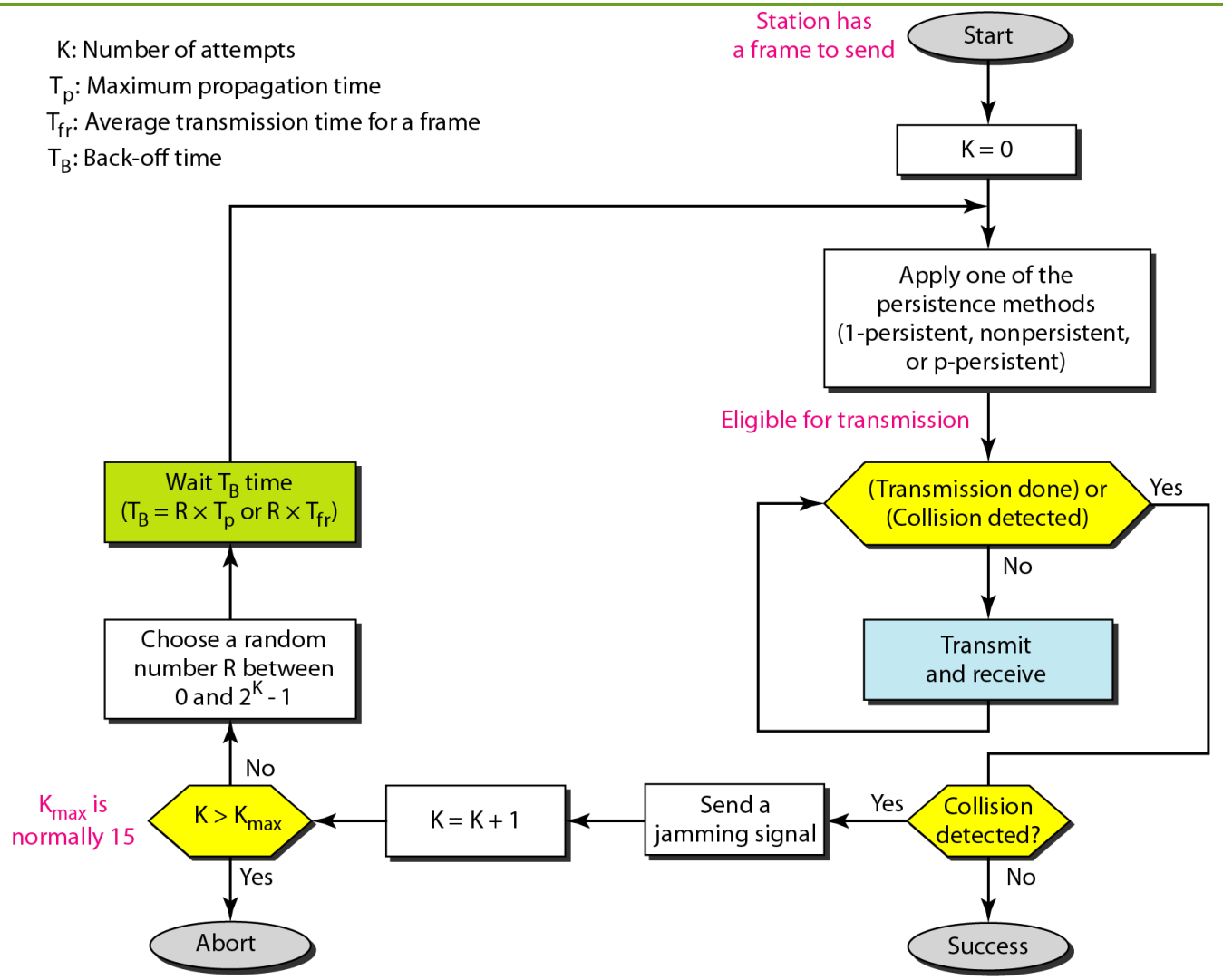


Figure 12.15 *Energy level during transmission, idleness, or collision*

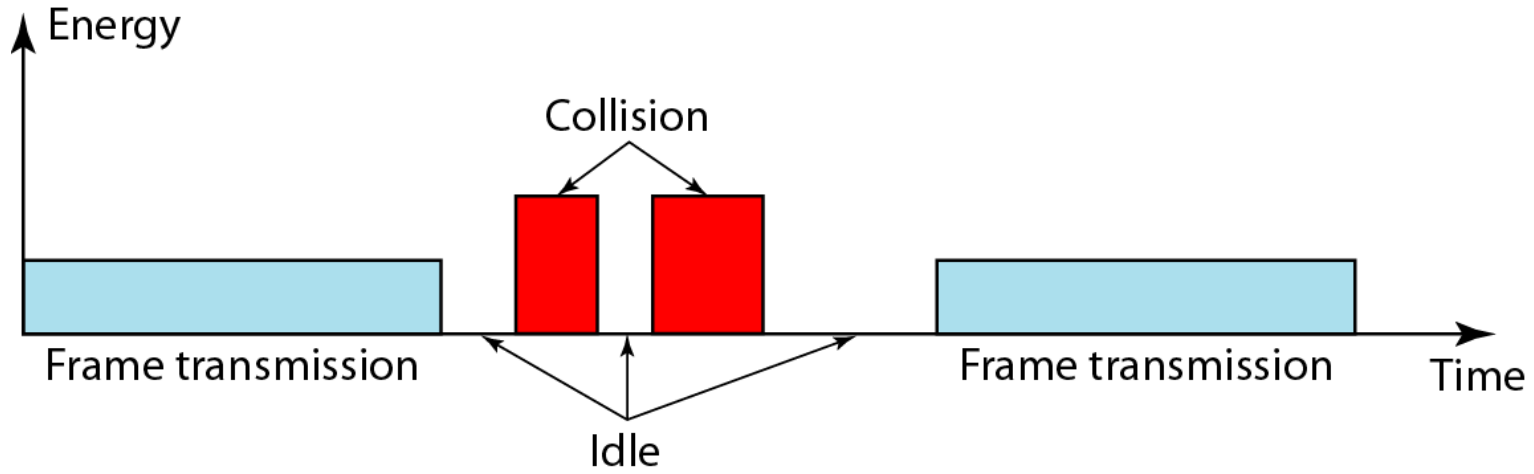
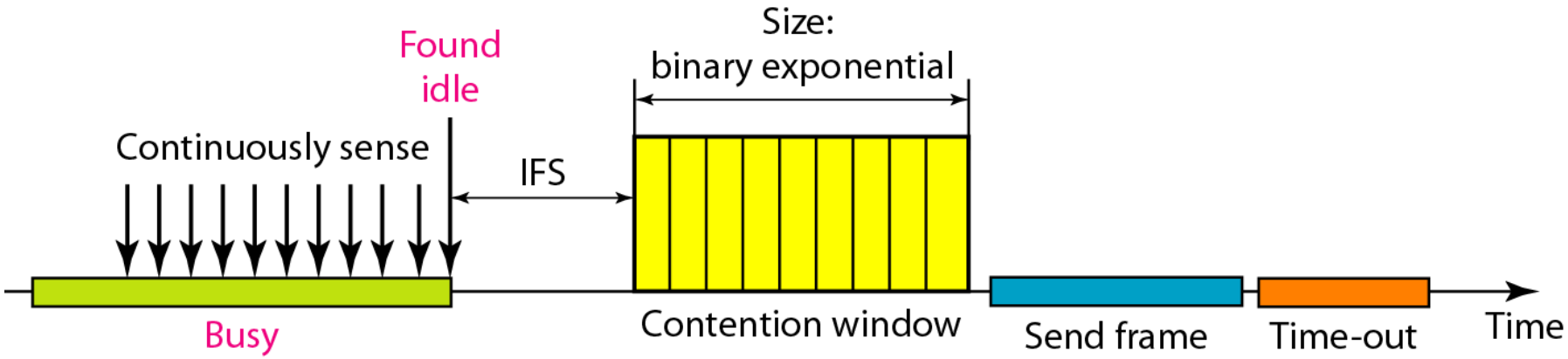


Figure 12.16 *Timing in CSMA/CA*



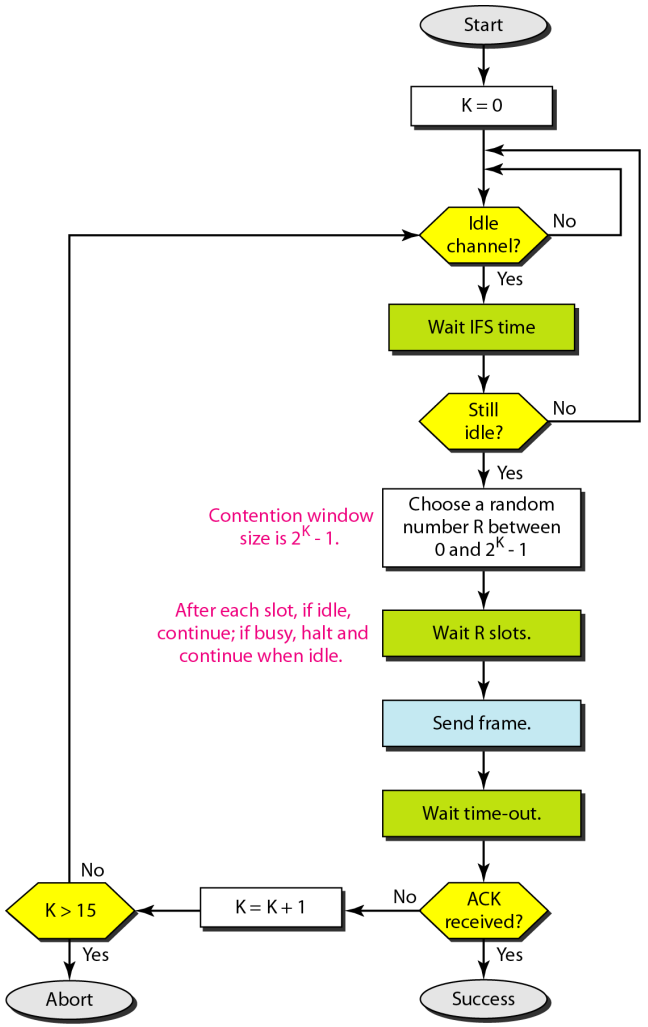
Note

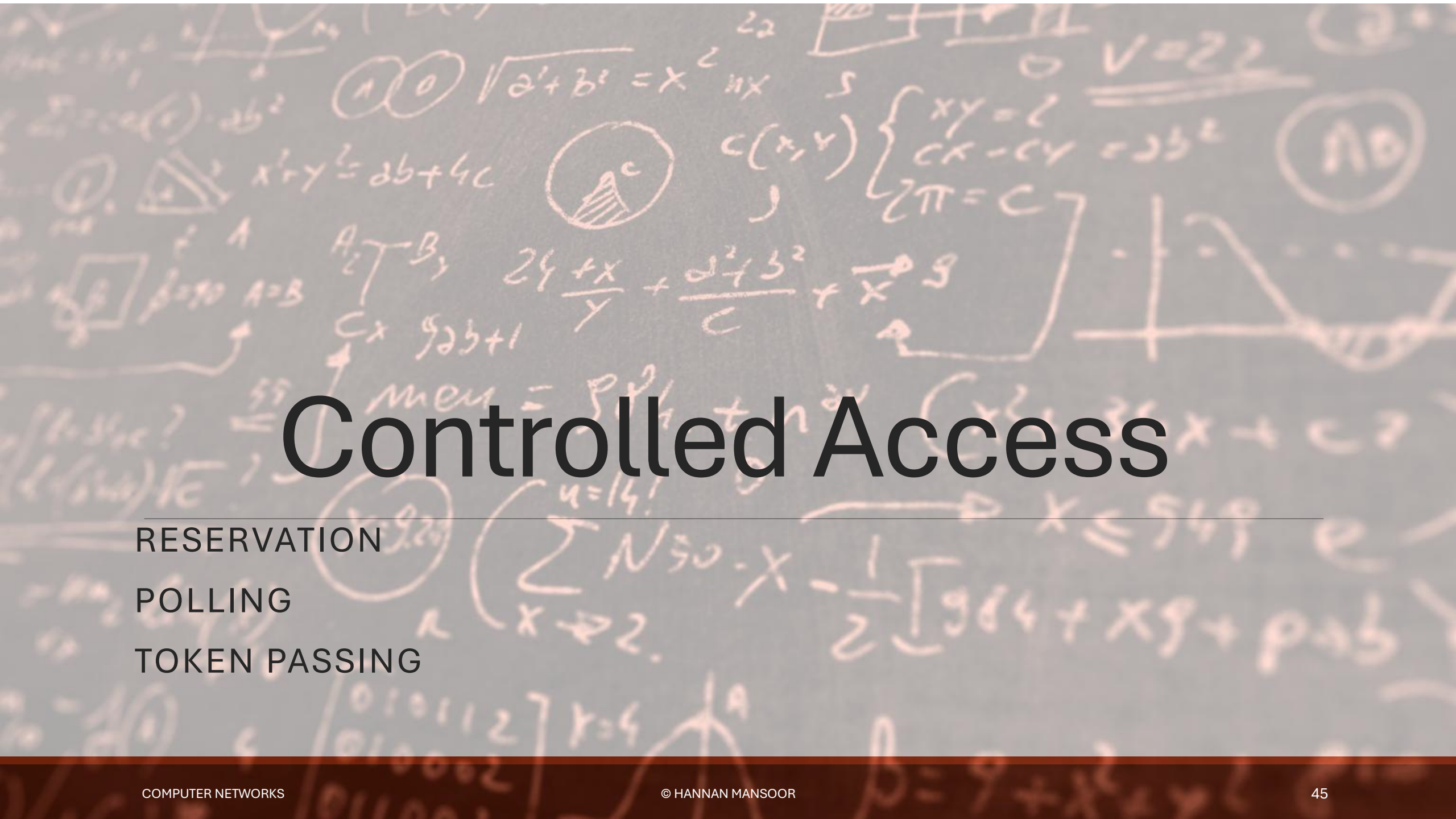
In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Note

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

Figure 12.17 *Flow diagram for CSMA/CA*





Controlled Access

RESERVATION

POLLING

TOKEN PASSING

© HANNAN MANSOOR

CONTROLLED ACCESS

- In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.

© HANNAN MANSOOR

Figure 12.18 *Reservation access method*

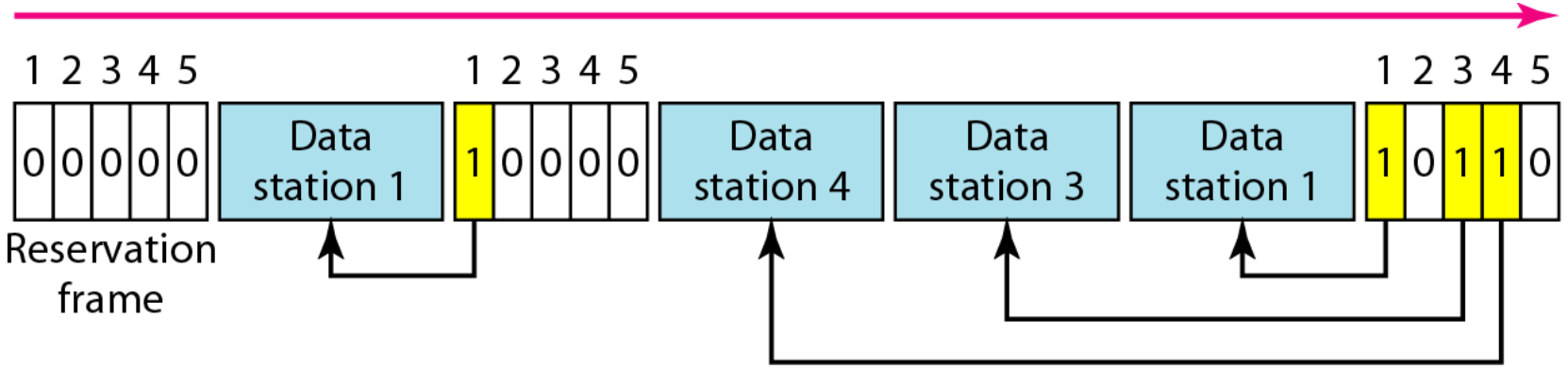


Figure 12.19 *Select and poll functions in polling access method*

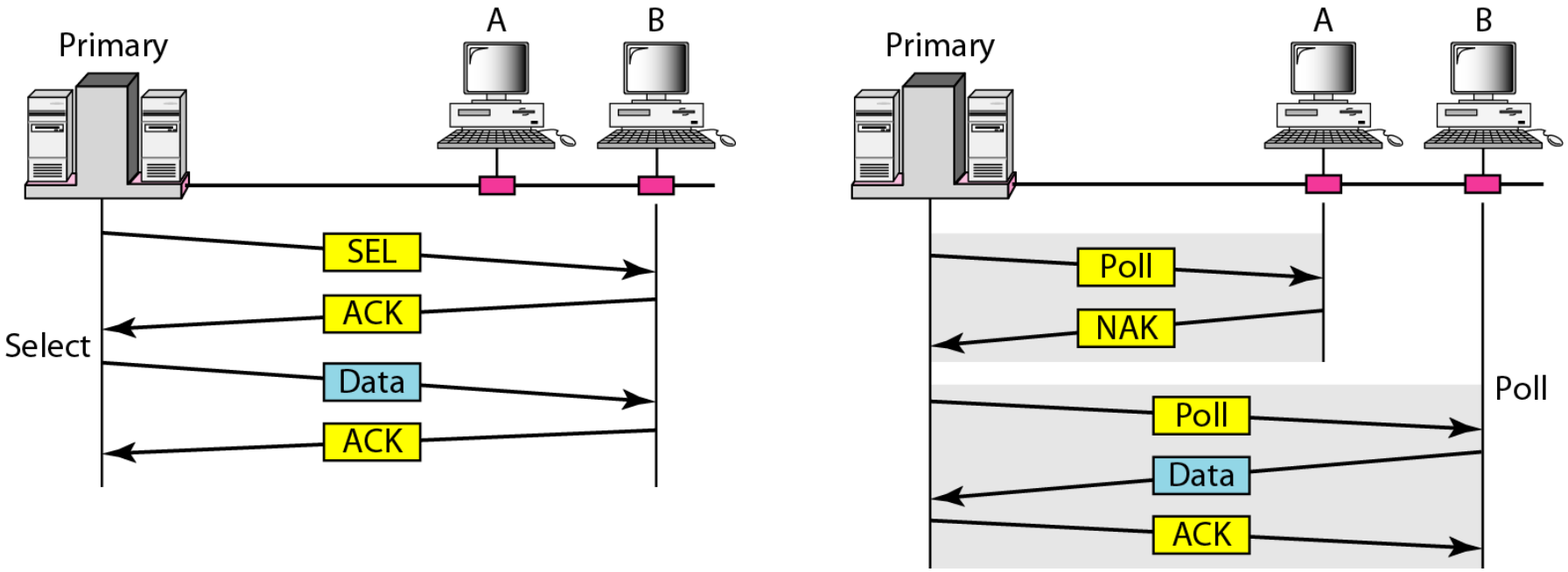
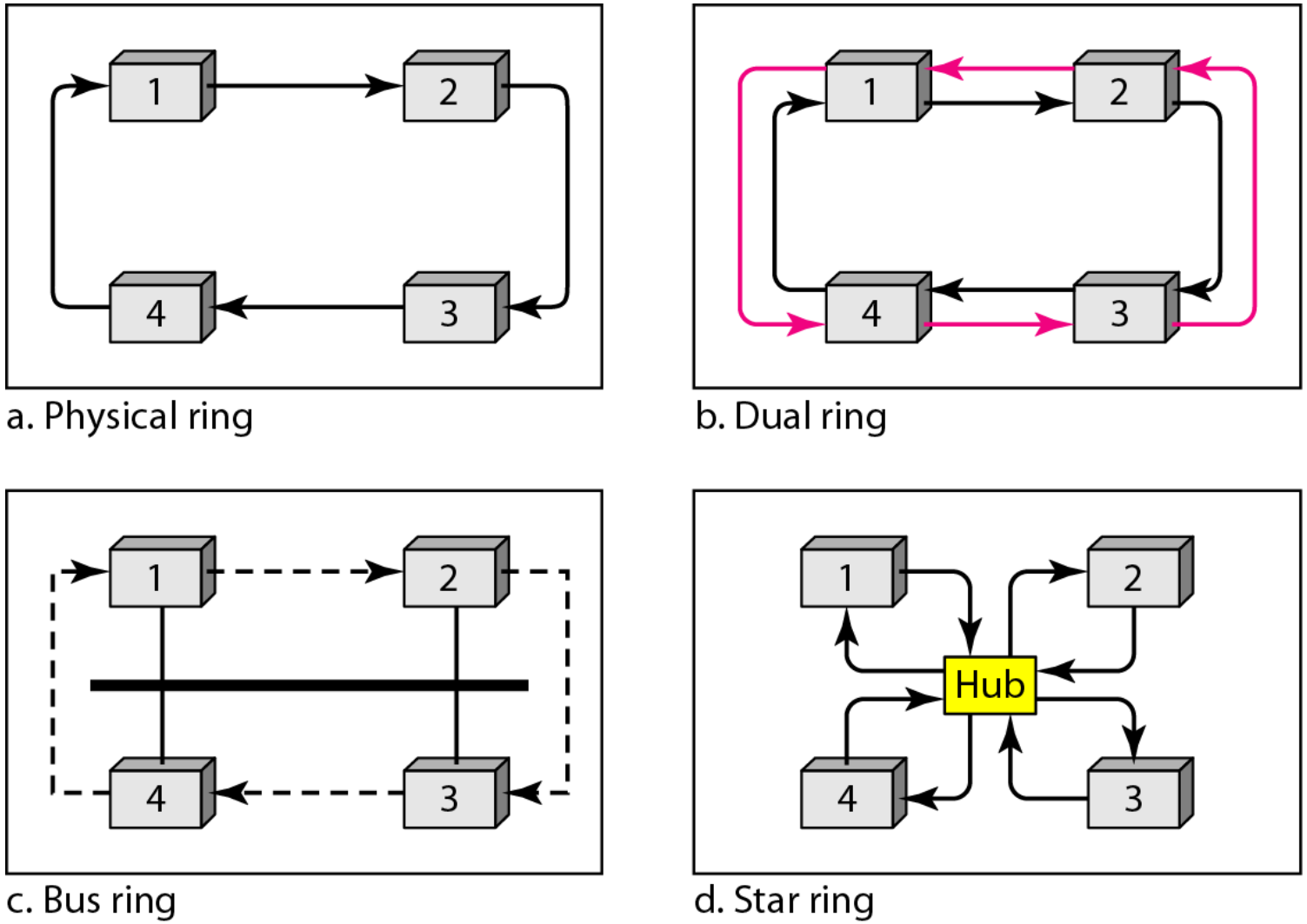


Figure 12.20 *Logical ring and physical topology in token-passing access method*





Channelization

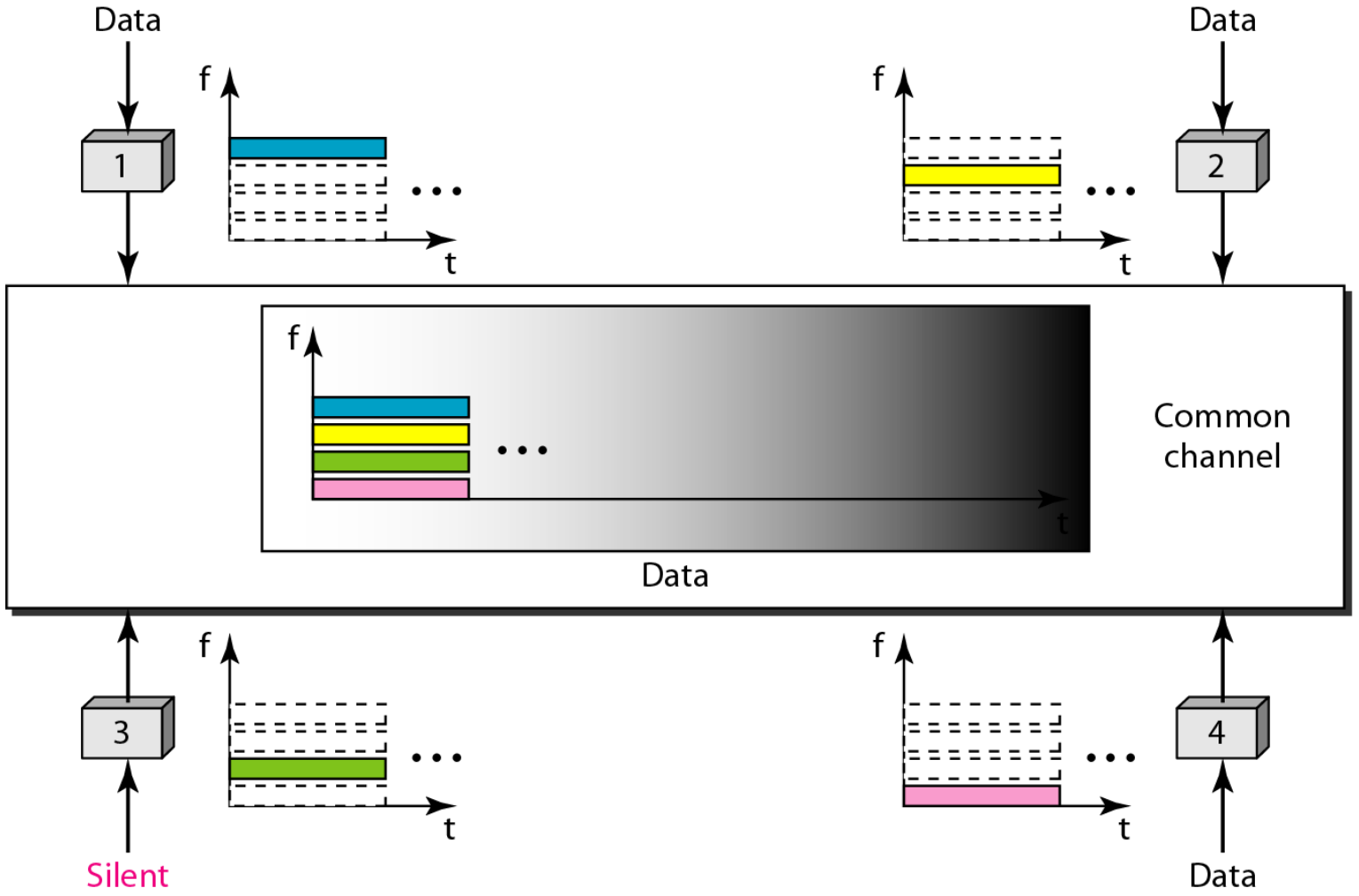
FREQUENCY DIVISION MULTIPLE ACCESS

TIME DIVISION MULTIPLE ACCESS

CODE DIVISION MULTIPLE ACCESS

© HANNAN MANSOOR

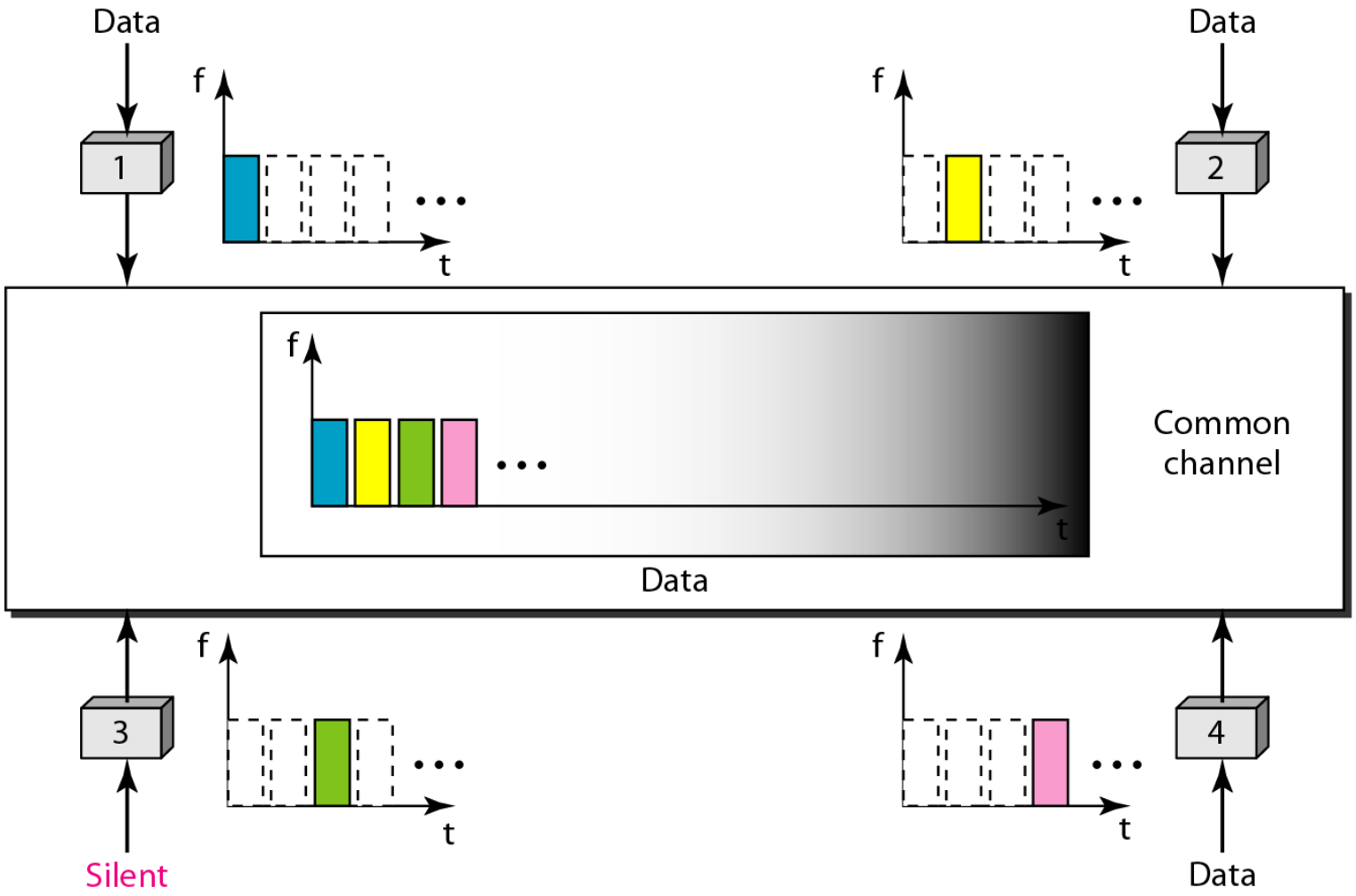
Figure 12.21 *Frequency-division multiple access (FDMA)*



Note

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

Figure 12.22 *Time-division multiple access (TDMA)*



Note

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

Note

In CDMA, one channel carries all transmissions simultaneously.

Figure 12.23 *Simple idea of communication with code*

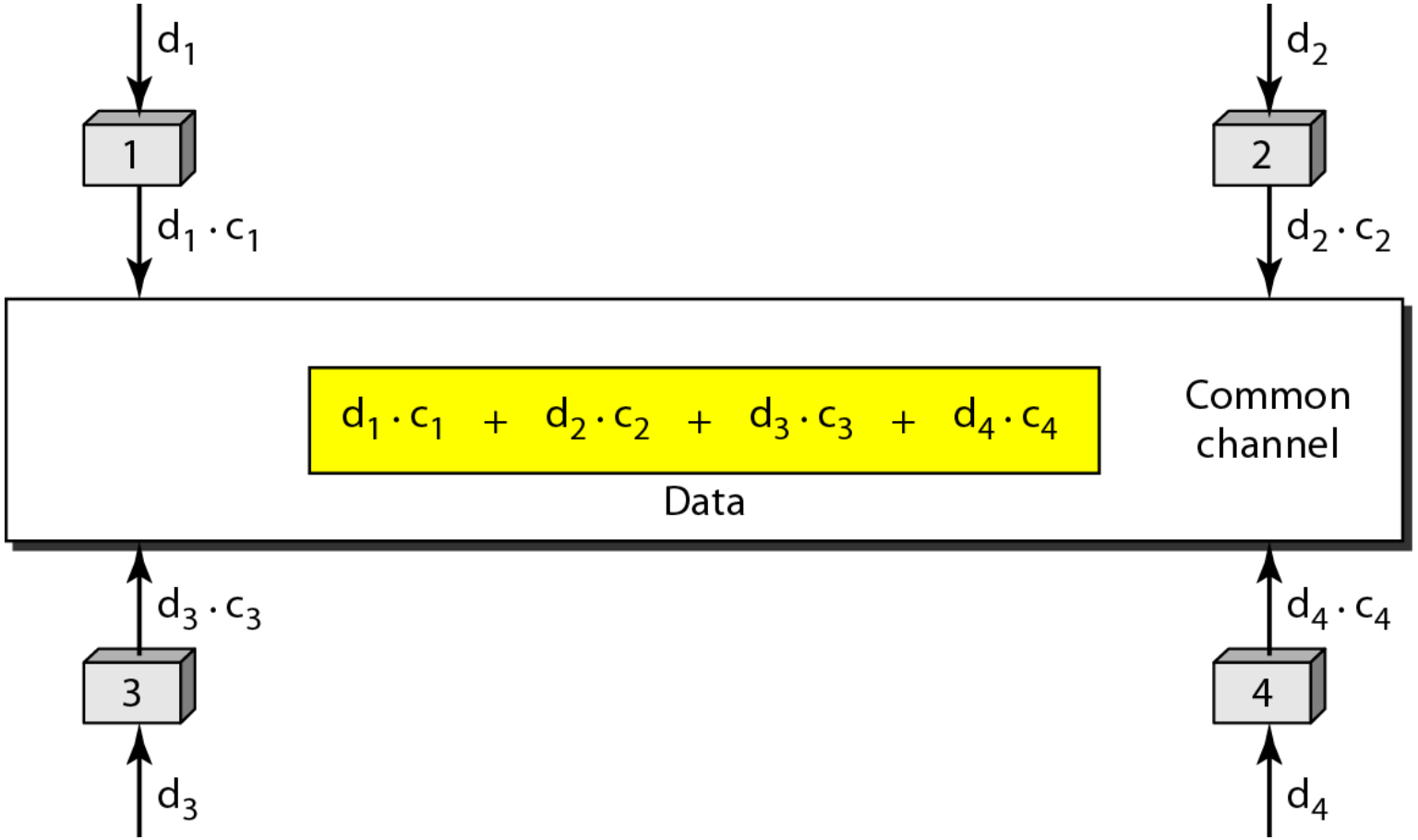


Figure 12.24 *Chip sequences*

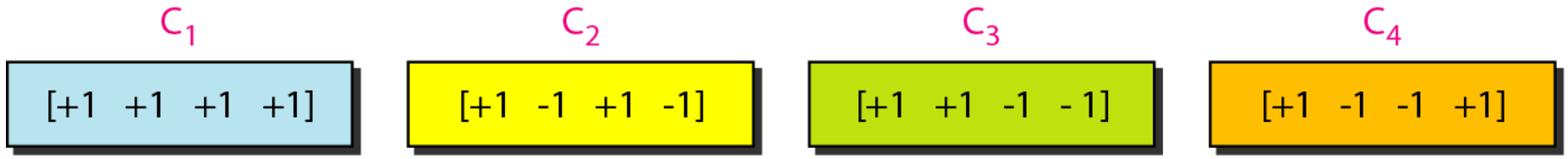
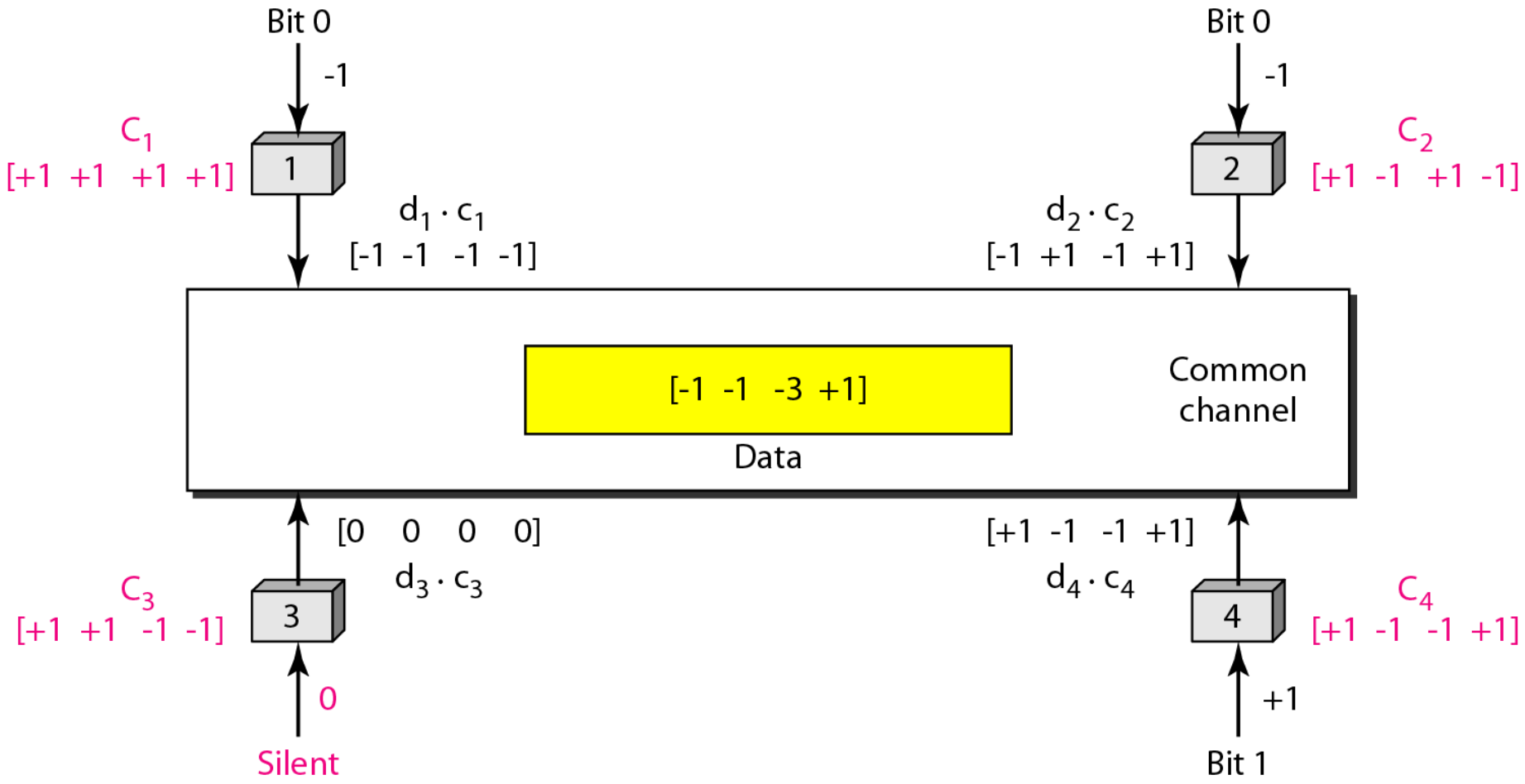


Figure 12.26 *Sharing channel in CDMA*



Wired LANs

IEEE STANDARDS FOR LANS AND MANS,

WIRELESS LAN: IEEE 802.11,

HIGH SPEED LANS.

© HANNAN MANSOOR



Blank Slide

© HANNAN MANSOOR



Refer Class Notes and Recommended Books for further Reading

Reference Books

- Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer networks* (5th ed.). Prentice Hall.
- Forouzan, B. A. (2012). *Data communications and networking* (4th ed.). McGraw-Hill.
- Kurose, J. F., & Ross, K. W. (2021). Computer networking: A top-down approach (8th ed.). Pearson.
- Peterson, L. L., & Davie, B. S. (2021). Computer networks: A systems approach (6th ed.). Morgan Kaufmann.
- Miller, M. A. (1992). Introduction to Digital and Data Communications. United States: West Publishing Company.
- Haykin, Simon, 3Rd Ed. (2008). Communication Systems India: Wiley India Pvt. Limited.
- Frenzel. (2016). Principles of Electronic Communication Systems. United Kingdom: McGraw-Hill Education.



Thank You